

Evaporation

Pre-Lab Plan
Unit Operations Lab #1
January 21st, 2020

Commented [DL1]: 29.5/35

Group 1
Tuesday PM

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1. Experimental Objective

The objective of this experiment is to concentrate a solution via a single-effect evaporator. The performance of the evaporator should be observed at different vacuum settings. Evaporation yields residue in viscous liquid form that allows for concentrating the solution. The vacuum increases the heat-transfer rate in the evaporator. The data will be utilized to calculate the capacity and economy at three different vacuum pressures.

Commented [DL2]: 4.5/5

Commented [AZ3]: The objective is your last sentence, that is using the single-effect evaporator to determine capacity and economy at 3 vacuum pressures. Also, you will be operating the evaporator at a constant concentration (that is what the make-up water is for).

2. Experimental

2.1 Apparatus

The main apparatus consists of a single effect evaporator composed of three main sections: the evaporator (heat exchanger), the condenser and the separator.

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A brief introduction of the apparatus is needed. You may include the flowsheet of the process.

Commented [AZ5]: The valve to add the makeup water and to drain the vessel are important components.

Commented [DL6]: Put this table below the picture.

Table 1: Apparatus Parts and Descriptions

No.	Instrument	Description
1	Flow Valve	Controls flow rate of water used to condense vapor
2	Pressure Valve	Controls vacuum pressure applied to system
3	Evaporation Vessel	Contains solution to be evaporated
4	Erlenmeyer Flasks	Recovers condensed steam
5	Heat Exchanger	Evaporates solution
6	Condenser	Condenses water vapor to liquid water.
7	Digital Temp. Reader	Displays temperature readings for 8 thermocouples throughout system
8	Temperature Gauge	Displays temperature of water before and after condenser.
9	Rotameter	Displays volumetric flow rate of the cooling water
10	Steam Trap	
11	Vacuum Pump	Adjust to obtain the desired pressure in the separator
12	Water Makeup Tank	Contains pure water to flow and maintain constant level
13	Steam Valve	Controls steam input to system located behind the apparatus (Not pictured)
14	Evaporator Pump Switch	Turns on and off the Evaporator Pump

Commented [DL7]: Unit. And other gauges should be mentioned.

Commented [DL8]: Unit and the maximum capacity.

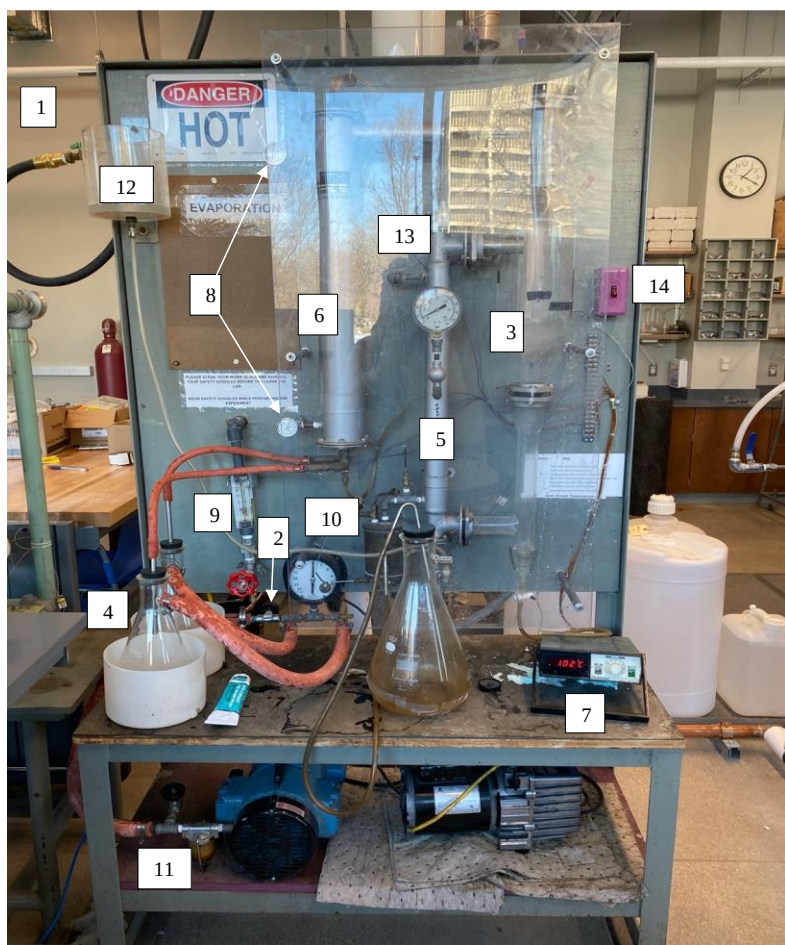


Figure 1: Labeled Evaporation Apparatus

Commented [AZ9]: Reflection makes the image hard to read.

2.2 Materials and Supplies

The materials and supplies that are required for the setup and the measurements of this experiment are listed in Table 2.2

Table 2.2: Materials and Supplies

Materials and Supplies	Purpose
Graduated cylinder	Collect steam condensate
Pure water	Help maintain steady state level from makeup tank and make sugar water solution
Stopwatch	To measure volumetric flow rate
Steam	Supplied to heat exchanger
Sugar	To make 10 wt% sugar water solution
Scale	Measure weight of the sugar

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Scoopula, vacuum grease, heat-resistance gloves, cooling water and the funnel are needed.

2.3 Experimental Plan (10 pts)

1. Prepare 10 wt% sugar solution and ensure contents of the separator are drained through the valve at the bottom
2. Fill the separator to the level line with the sugar solution allowing some to flow into the draining tube
3. Add pure water in the makeup tank
4. Start the steam flow
5. Start the cooling water to the steam trap and collect the condensate in a graduated cylinder
6. Turn on the vacuum pump
 - a. Adjust bypass valve to a desired pressure (reads in inches of mercury)
 - b. Adjust the flow of the vacuum until it becomes steady
7. Collect vapor product being conscious that the concentration of the solution will not change
8. Allow the system to reach steady state (based on the temperature readings)
9. Record the level of the steam condensate and the collected vapor.
10. Wait 15 minutes; record the time and the levels again.
11. Calculate the flowrates of the two streams.
12. Record the inlet and outlet temperature and the flowrate of cooling water.
13. Determine how much steam has condensed in the exchanger.
14. Change the pump to a different vacuum level two more times.

Commented [DL11]: 8/10

Commented [DL12]: How much?

Commented [DL13]: Which one should be the first?

Commented [AZ14]: Cooling water feeds steam and condenser. Water should also be going to the condenser and adjusted with the flowmeter.

Commented [DL15]: How much?

Commented [DL16]: Have you noticed that the left side and the right side of the gauge have different units?

Commented [AZ17]: You may need to secure the valve on the vacuum line to maintain a constant vacuum pressure, it tends to vibrate and loosen.

Commented [DL18]: Good.

Commented [DL19]: Be specific.

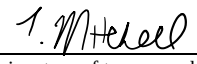
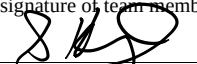
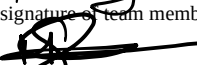
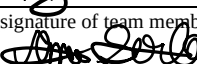
Project Safety Evaluation (10 pts)

Project Title: Evaporation

Date: January 21st, 2020

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Filling vessel near exposed wire for thermocouple setup to digital display for temperature readings.
Potential mechanical hazards?	Yes	If yes, identify conditions: Vacuum pressure with compromised and/or cracked glass could cause apparatus to implode.
Potential pressure hazards?	Yes	If yes, identify conditions: Pressurizing system with closed valves
Potential temperature hazards?	Yes	If yes, identify conditions: High temperature steam at approximately 105 °C.
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	Yes	If yes, specify type: As a precaution, heat-resistant gloves may be worn when starting steam flow into system.
MSDS reviewed by all team members?		List MSDS sheets reviewed: Sucrose, Water [Steam]
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	110	Heat exchanger and piping containing steam
Pressure (psig)	15	Heat exchanger and piping containing steam
Safety Controls	yes/no	If yes, Provide description

I have read relevant background material for the Unit Operations Laboratory entitled: “ Evaporation ” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.


 signature of team member January 21st, 2020
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Commented [DL20]: 9/10

Commented [AZ21]: Might be slightly higher.

Commented [DL22]: List the CAS number.

Commented [AZ23]: What about cooling water pressure and temp?

Commented [DL24]: About 120 degree Celsius.

Commented [DL25]: 2-15 psia.

Commented [DL26]: Steam inlet from steam valve to regulate the amount of steam coming in system.
Thermometer on the heat exchanger to make sure safe temperature of heat.
Pump plug to regulate amount pressure of system - can be quickly turned on/off.
Main power switch can be used to cut power from all the electrical equipment.